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RESEARCH ARTICLE

An optimization model for the real-time truck dispatching problem in open-pit mining operations

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Abstract

This work is concerned with the real-time truck dispatching problem in open-pit mining operations. The proposed methodology is built on a multi-stage dispatching approach to optimize the truck-shovel material handling systems. The procedure consists of two main stages: allocation planning and dynamic allocation. At first, a scenario-based method is presented to estimate the optimal size of trucks in the mining operation. Then, a novel multi-objective mathematical model for solving the problem of dynamically allocating trucks is presented, aiming at minimizing fleet waiting times and deviations from the path production requirements established by the allocation planning. To evaluate the performance of the proposed model, three different heuristic methods, including criteria of minimizing shovel idle time, minimizing ratio variance, and minimizing the deviation from the allocation planning, are developed. Afterward, the dispatching methods are tested by building a discrete-event simulation of a copper ore mine case study to compare the results. The findings suggest that the multi-objective model developed in this paper has a great potential to meet the production requirements of the operation compared to the heuristic algorithms.

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1 Introduction

Truck dispatching management is one of the significant issues in open-pit mining operations because of the expenditures on material transportation, which can exceed half of the total operating costs. Basically, real-time truck dispatching decision is a combinatorial problem that involves finding the optimal dynamic assignment of trucks to shovels while enhancing equipment efficiency and reducing haulage costs. Making poor dispatching choices, which lead to long queues at service stations, can have a hugely negative impact on transportation performance. To solve this problem, fleet management systems (FMSs) can be used to optimize mining operations. An efficient FMS as very short-term planning can result in an increase in operating profits and a decrease in the required truck fleet size, as well as an increase in production of open-pit mines.

Following the literature on truck dispatch systems, three types of strategies are used for assigning a truck to an appropriate shovel. The first strategy, called 1-truck-for-n-shovels, considers n possible shovels for a truck waiting for an assignment. This strategy ignores the impact of the real-time decisions taken in the next truck dispatching request. All previous studies (Chatterjee and Brake 1981; Tu and Hucka 1985; Lizotte and Bonates 1987; Sadler 1988; Li 1990; Forsman et al. 1993; Xi and Yegulalp 1993; Ataeepour and Baafi 1999) have been conducted using heuristic methods that do not guarantee satisfactory solutions under all possible conditions. Collectively, the main weakness of these studies is the failure to address mathematical optimization models for achieving exact solutions. Moreover, the existing methods fail to resolve the contradiction between multi objectives in the mining operation. The second strategy is called m-trucks-for-1-shovel, where the truck dispatching decisions are made by considering one shovel at a time and m trucks that will request dispatch in the near future. In the literature, the application of this strategy for making dispatching decisions has been excellently reviewed by White et al. (1982); White and Olson (1986); Soumis et al. (1989); Moradi-Afrapoli et al. (2019a); Moradi-Afrapoli et al. (2019b); Mohtasham et al. (2022). Finally, in the m-trucks-for-n-shovels strategy, m trucks, which have to be dispatched in the near future, are assigned to n shovels, allowing a more global view of the problem. Implementation of such a dispatching strategy can be found in Elbrond and Soumis (1987); Temeng et al. (1997); Subtil et al. (2011); Alexandre et al. (2019); Yeganejou et al. (2021); Smith et al. (2021).

According to Alarie and Gamache 2002, the truck dispatching approaches can be grouped into single-stage and multi-stage methods. The single-stage approaches dispatch the trucks using heuristic procedures without considering any constraint or production target, usually following some rules of thumb. In the last decades, some studies have been conducted on the single-stage approaches to support truck dispatching decisions in open-pit mines based on heuristic processes (Forsman et al. 1993; Arelovich et al. 2010; Jaoua et al. 2012; Bastos 2013; Hashemi and Sattarvand

2015; Cox et al. 2017; Dumakor et al. 2017; Koryagin and Voronov 2017; Ahumada et al. 2020). The main weakness of this method is the failure to address the production schedule, optimal rates of the shovels, mine history, maintenance plan, and processing capacity. The multi-stage method divides the problem into two main sub-problems, including truck and shovel allocation optimization in an upper stage and real-time truck assignment optimization in a lower stage. The first stage solves the fleet allocation problem by creating a fixed schedule that satisfies given production requirements. To do this, a programming model is employed considering the objectives of the short-term production scheduling in the mining operation. Most studies provide single-objective mathematical models to solve the fleet allocation plan in open-pit mines, such as linear programming (LP) (White and Olson 1986; Li 1990; Gurgur et al. 2011; Subtil et al. 2011), non-linear programming (NLP) (Soumis et al. 1989; Soumis et al. 1990; Ta et al. 2005; Sahoo et al. 2010), mixed-integer programming (MIP) (Eivazy and Askari-Nasab 2012; L'Heureux et al. 2013; Chang et al. 2015; Torkamani and Askari-Nasab 2015; Bajany et al. 2017), and stochastic programming (SP) (Ta et al. 2005; Topal and Ramazan 2012; Matamoros and Dimitrakopoulos 2016; Bakhtavar and Mahmoudi 2020). The implementation of the above models results in either determining the number of truck trips required to meet each path's desired production rate or the total tonnage to be moved from different mining faces to the destinations in the mine. As regards the heterogeneous equipment fleet used in the vast majority of mines, perhaps the most serious disadvantage of these models is that they fail to take into account compatibility constraints between trucks and shovels. Another problem with these models is that they ignore the effect of all other objectives involved in open-pit mining operations. Another shortcoming of this method can be found in the shovel allocation problem as a link between the production operations and the mine strategic plans. These limitations can drive the final results far from the real optimal values. However, there are, to date, limited published studies overcoming the abovementioned drawbacks of the fleet allocation problem. In fact, few researchers (Temeng et al. 1989; Upadhyay and Askari-Nasab 2016; Mohtasham et al. 2017; Manríquez et al. 2019; Both and Dimitrakopoulos 2020; Mohtasham et al. 2021a) have focused on multi-objective models to meet the production requirements in the mine operation level, which deal with multiple competing objectives. The approach suggested in this study is based on a mixed-integer linear goal programming (MILGP) model proposed by Mohtasham et al. (2021b) for the upper stage of multi-stage fleet management systems. The model presents a more complete mathematical programming model of truck-shovel allocation planning than those found in the literature. The second stage is the actual real-time dispatching, which is implemented to minimize the deviation from the plan generated by the first stage. This part uses innovative approaches or mathematical models for the dynamic assignment of the trucks to the shovels. Heuristic methods have been quite popular in the literature because they are easy to use and do not require much computation when making dispatching decisions in real-time. White and Olson (1986) present an algorithm for assigning the best truck to the most demanding shovel that is the basis for the DISPATCH® software developed by the Modular Mining Systems. Soumis et al. (1989) use an assignment problem to minimize the sum of the squared deviation of the estimated waiting time of trucks from the planned waiting time. Sgurev et al. (1989) describe an automated system, called TRASY, for real-time control of the truck haulage to minimize the degree of saturation of each path in open-pit mines. Li (1990) introduces a truck dispatching algorithm based on the maximum inter-truck-time deviation rule to keep truck flows as close to the optimum as possible. Xi and Yegulalp (1993) implement a real-time dispatch algorithm based on the concept of a "minimizing ratio variance" rule. Temeng et al. (1997) propose a transportation algorithm, which minimizes the cumulative deviations of the optimal production target for each shovel path. Subtil et al. (2011) apply a dynamic dispatching heuristic used in the commercial package SmartMine that runs an exhaustive search over upcoming decisions. Ahangaran et al. (2012) develop a MILP model to solve the dynamic truck assignment problem. The model minimizes the total cost of loading and transportation according to the operational constraints of the mine. Moradi-Afrapoli et al. (2019) provide a transportation model based on a MILP model to simultaneously minimize shovel idle times, truck wait times, and deviations from the path production requirements. Alexandre et al. (2019) create solutions to the real-time truck dispatching problem using a multi-objective mathematical model in an open-pit mine. The aims of the model are simultaneously maximizing the total production and minimizing costs under some operational and physical constraints. Smith et al. (2021) develop two policies for real-time truck dispatching decisions in an open-pit mine. The first dispatching policy renders an operational plan using a non-linear optimization model that considers queueing effects to set target average flow rates between mine locations. The second policy provides a formulation based on a mixed-integer programming model for optimal decision-making on the real-time truck dispatching problem. Mohtasham et al. (2022) propose a heuristic algorithm to dispatch trucks to shovels in a multi-stage fleet management system. The heuristic algorithm considers two objectives, namely the minimization of shovel idle times and deviations from the path production requirements established by the upper stage. Moreover, other algorithms regarding the dynamic allocation of trucks can be found in the studies by He et al. (2010), Souza et al. (2010), Tan et al. (2012), Patterson et al. (2017), Ozdemir and Kumral (2019), Samavati et al. (2019), Seiler et al. (2020). Most real-time truck dispatching systems use the multi-stage approach because it considers production targets. The focus of this work is to address the lower stage of this problem, as a limited number of methods can be found in the literature that deals with the truck dispatching problem. The dynamic truck allocation model described in this research differs from others in the literature in terms of objective functions and imposed constraints in the mine.

Simulation involves designing a model of a real-world process or system over time and then performing experiments with the model for the purpose of understanding the actual behavior of the system and evaluating various types of methods for system performance. With regards to this, the researchers evaluate the performance of their proposed methods for real-time control decisions using discrete-event simulation. Figure 1 presents the control loop of the multi-stage FMSs in the literature of the academic group. The systems require continuous monitoring of the locations of trucks and shovels. At first, they receive all the necessary input data from the historical dispatching database, and the upper stage is run at the beginning of the shift and during the shift when the input parameters change due to specific circumstances such as the

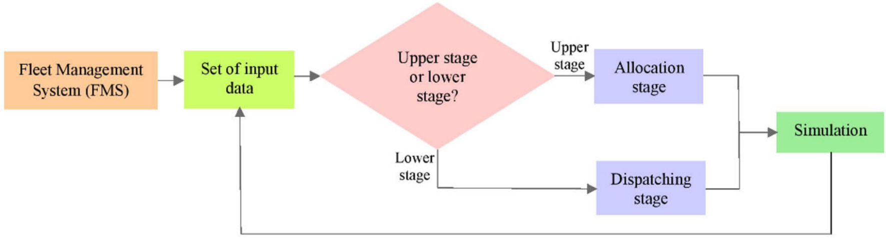

Fig. 1 The control loop of the multi-stage FMSs in open-pit mines

failure of the truck-shovel fleet to generate new set-up production rates of the paths. The lower stage within the simulation model is called using the real-time monitoring information when the truck operator requests a dispatch. This process is performed repeatedly until the shift ends. To do this, in this research, we link the simulation model to an optimization model for real-time truck dispatching decisions to improve equipment utilization, increase productivity, and reduce waiting times of the mining equipment.

Despite the extensive research on the fleet allocation schedule, the real-time truck dispatching stage has not attracted enough attention, and there is more that could be done. Furthermore, what is not yet clear is the impact of applying mathematical models on the 1-truck-to-n-shovels strategy. In other words, to what extent the difference between methods can be affected the total mine productivity and equipment utilization remains to be established and will require mathematical modeling. Another drawback of the presented methods for this strategy is that there are given no-decision between various conflicting objectives in the mining transportation system. However, in the real world, the mining operation is complex, and optimal dispatching decisions need to be taken in the presence of multiple objectives. Therefore, there are two primary purposes of this research: 1. To develop a multi-objective goal programming model based on the 1-truck-to-n-shovels strategy for the real-time truck dispatching decision that aims at finding the best trade-off among the main objectives of minimizing the deviations from optimal production rates of the paths and minimizing the truck and shovel waiting times in the operation. The presented model can be easily adapted to be applied in other open-pit mines with various operational conditions. 2. To compare the proposed model against the relevant heuristic algorithms from the literature using a discrete-event simulator. The model is analyzed with three diverse heuristic methods, including criteria of minimizing shovel idle time, minimizing ratio variance, and minimizing the deviations from the fleet allocation plan in actual real-time truck dispatching decisions. The model is shown to outperform other algorithms, and it has been proven that the heuristics do not guarantee the optimal solutions. The most basic contribution of this work is to compare the heuristic algorithms against the proposed mathematical model for the destination management of trucks.

In the following sections, the research methodology for this study is presented first, along with details of the mathematical formulations. Then, we present a description of how the simulation–optimization framework works. In the fourth section, an open-pit copper mine is introduced as a case study to validate the proposed approaches. The next section of the article is focused on the results of using our developed truck dispatching model and compares them against the results of using the heuristic algorithms. Finally, the last section presents the conclusions and recommendations for future research.

2 Dynamic truck allocation methodology

There is no research on the comparison of mathematical and heuristic methods for the real-time truck dispatching problem based on the 1-truck-to-n-shovels strategy. Hence three heuristic algorithms and one mathematical model are investigated to solve the truck dispatch problem in an open-pit mine. Each of these methods has a specific policy for real-time decision-making. The dynamic dispatch methods described in this study are classified as a multi-stage dispatch approach, as shown in Fig. 2. It has two main stages: allocation planning and dynamic allocation. The first step, which is a major component in the multi-stage methods, deals with a fleet allocation schedule to determine the optimal amount of materials for each path and assign trucks and shovels to the mining faces. This step is vital in this respect that the real-time dispatching decisions are affected by considering the results of the fleet allocation schedule. According to the literature, the fleet allocation model developed by Mohtasham et al. (2021b) is rich enough to apply in the first stage. So, this model is chosen for the upper stage in the presented multi-stage decision-making for the truck dispatching. The model is a multi-objective model based upon a MILGP formulation framework. It determines the optimal production plan of the shovels and allocation plan of the trucks and shovels with the aim of maximizing production, meeting desired head grade and tonnage at the ore destinations, and minimizing fuel consumption of trucks under operational constraints. The model is solved at the beginning of each shift or

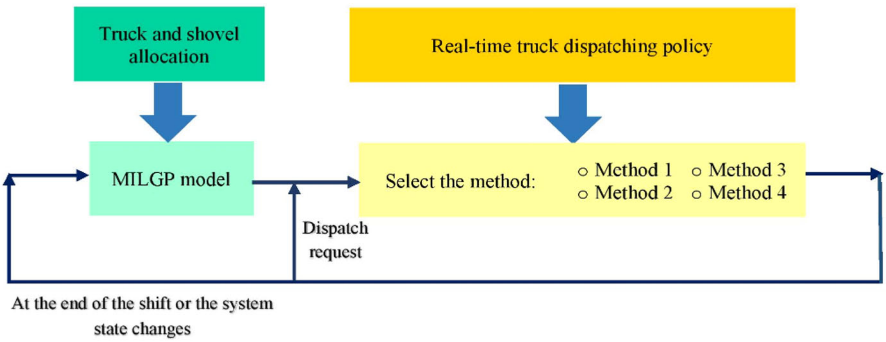

Fig. 2 The framework of the proposed multi-stage approach for the truck dispatching problem

when the system state changes. More details about the model can be found in the study of Mohtasham et al. (2021b). All the solutions obtained from the upper stage are converted into input data for the lower stage (real-time dispatching stage) to find the optimal decisions according to diverse truck dispatching policies. In the specific dispatching part of the proposed framework, two approaches are set: the mathematical optimization model and the heuristic optimization algorithms. It is noteworthy that all dispatch methods in this study deal with the 1-truck-to-n-shovels strategy based on a multi-stage approach. The well-known heuristic algorithms are used as a comparison baseline for the developed model. This is due to the fact that there are no previous studies investigating the development of a mathematical model using this strategy. The presented mathematical model, called Method 1, includes two main objective functions trying to simultaneously minimize shovel idle times, truck waiting times, and deviation from the path production requirement established by the upper stage. The heuristics algorithms for the real-time truck dispatching are Method 2 with minimizing shovel idle time, Method 3 with minimizing ratio variance, and Method 4 with minimizing the deviation from the fleet allocation plan. The dispatch process approach is called every time a new truck assignment is required. The applied approaches for real-time decisions are described in detail in the next subsections.

2.1 Method 1: a mathematical optimization model

In practical problems, especially in the mining industry, optimal decisions need to be taken with regard to the presence of trade-offs between several conflicting objectives. Although various approaches address the truck dispatching problem in open-pit mines, these methods ignore the multiple objectives in mining operations. Therefore, in this research, a multi-objective model to dispatch trucks to shovels in real-time is developed based on the 1-truck-for-n-shovels strategy at any time shift. The proposed approach can overcome the imperfections and limitations of the previous studies. The most important limitations are as follows:

- Failure to achieve exact solutions
- Failure to guarantee satisfactory solutions
- Failure to multi-objective optimization

The major benefit of the method is that the truck dispatching decisions are optimally made using a mathematical programming model. Another advantage of the method lies in the optimization of more than one objective function simultaneously. Before formulating the truck dispatching model, the following notations are defined.

The following indexes are identified:

t Index for the truck requesting dispatch

d Index for the current destination of truck t

s Index for a set of shovels: $s \in \{1, \ldots, S\}$

The parameters used in the model are as follows:

$T_{arrival[t,\mathrm{d,\,s}]}$	The arrival time of empty truck t from the current destination d to shovel s (min)
$S_{available[\mathrm{t,\,s}]}$	The next time shovel s is available to serve truck t (min)
$C_{truck[t]}$	The capacity of the truck t (ton)
$P_{current[\mathrm{d,\,s}]}$	The current path production rate for the path from the current destination d to shovel s (ton)
$P_{target[\mathrm{d,\,s}]}$	The target path production rate for the path from the current destination d to shovel s based on upper stage decision (ton)
$\gamma_{[\mathrm{t,\,s}]}$	Binary parameter indicating whether the truck t is compatible with shovel s or not

The decision variables are:

$x_{[t,\,\mathrm{d,\,s}]}$	Binary integer variable to assign truck t from the current destination d to shovel s
$c^-_{[\mathrm{d,\,s}]}$	Negative deviation from the target path production rate for the path between the current destination d and shovel s
$c^+_{[\mathrm{d,\,s}]}$	Positive deviation from the target path production rate for the path between the current destination d and shovel s

The mathematical model for this optimization problem is expressed with Eq. (1)–(8). In the model formulation, $T_{arrival[t,\mathrm{d,\,s}]}$ and $S_{available[\mathrm{t,\,s}]}$ are defined as stochastic parameters by the distribution functions to capture the uncertainty of the mining transportation operation.

$$f_1 : min \sum_s \left| T_{arrival[t,\mathrm{d,\,s}]} - S_{available[\mathrm{t,\,s}]} \right| \times x_{[t,\,\mathrm{d,\,s}]} \tag{1}$$

$$f_2 : min \sum_s \left(c^-_{[\mathrm{d,\,s}]} + c^+_{[\mathrm{d,\,s}]}\right) \tag{2}$$

Subject to:

$$C_{truck[t]} \times x_{[\mathrm{t,\,d,\,s}]} + P_{current[\mathrm{d,\,s}]} + c^-_{[\mathrm{d,\,s}]} - c^+_{[\mathrm{d,\,s}]} = P_{target[\mathrm{d,\,s}]} \; \forall s \in \{1, \ldots, S\} \tag{3}$$

$$\sum_s x_{[t,\,\mathrm{d,\,s}]} \le 1 \tag{4}$$

$$x_{[t,\,\mathrm{d,\,s}]} \le \gamma_{[\mathrm{t,\,s}]} \; \forall s \in \{1, \ldots, S\} \tag{5}$$

$$x_{[t,\,\mathrm{d},\,\mathrm{s}]} \in \{0, 1\}\, \forall s \in \{1, \ldots, S\} \tag{6}$$

$$c^{-}_{[\mathrm{d},\,\mathrm{s}]} \geq 0\, \forall s \in \{1, \ldots, S\} \tag{7}$$

$$c^{+}_{[\mathrm{d},\,\mathrm{s}]} \geq 0 \forall s \in \{1, \ldots, S\} \tag{8}$$

The presented model involves two different objectives. The first objective function, presented in Eq. (1), minimizes the truck or shovel idle times in material handling. The second objective function, presented in Eq. (2), is a goal programming objective that minimizes the deviation from the production rate of the paths. It should be noted that the objective function (1) has a higher priority than the second one in the considered dispatch policy. The constraints define key aspects of the truck dispatching in the mining operation. Constraint (3) computes the deviation of the path production rate for each path connecting the current truck t position to a shovel from the desired path production rate. This allows the model to understand how much the shovels are behind their specific target. Constraint (4) avoids the assignment of truck t to multiple shovels in the decision-making process. This constraint is active when there is the same priority for the shovels to dispatch the truck. Constraint (5) is related to the compatibility between the truck and shovels to avoid dispatching a truck to a mining face with no compatible shovel. Generally, there are two criteria for determining the compatibility between trucks and shovels, making a substantial contribution to improving the fleet's efficiency: (1) shovel's operator vision, and (2) the number of a shovel's passes to fully load each truck type. These criteria mean that a truck cannot be dispatched to incompatible shovels. Positioning the shovel at a suitable height is a good common practice for a better vision of the shovel's operator. Three to five passes per truck payload are suitable for wheel loaders, whereas five to seven bucket loads are acceptable for shovels. Finally, constraints (6–8) are non-negatively constraints for the decision variables.

The developed model has two objectives with different dimensions which cannot be optimized simultaneously. In such models, trade-off decisions are required to achieve the best compromise solution. The proposed model is solved using Sequential Linear Goal Programming (SLGP) procedure, which is a widely used approach for solving programming problems with multiple objective functions. In this solution approach, various objective functions are first ranked by the decision-maker considering their priority level in order from the most important to the least important. Then, the model is solved using a sequence of conventional LP models. The first optimization process begins with optimizing the objective function that has the highest priority. The process continues in a sequence of decreasing priority until the problem is completely resolved. Considering the solution of the previous LP, an augmented constraint is added to the next models. More detailed information regarding this solution approach can be found in Mohtasham et al. (2021b). Figure 3 shows the process of solving the model based on the SLGP approach in detail. As mentioned earlier, in this research, it is assumed that objective function 1 has a higher priority than another one.

According to the proposed model structure used with the 1-truck-to-n-shovels strategy, the model can be solved in any size of the mining operation by the available

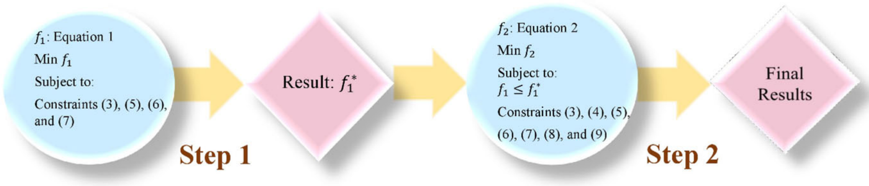

Fig. 3 The solving process of the presented model

optimizer software. The average typical runtime for solving the model in the GAMS software environment with the CPLEX solver is about 1 s.

2.2 Heuristic optimization algorithms

This section presents three heuristic algorithms for real-time truck dispatching, which are considered benchmarks for assessing the performance of the proposed mathematical model. There are two main reasons for choosing these heuristic methods: (1) no mathematical model has yet been developed based on the 1-truck-to-n-shovels strategy, and (2) these methods are well-known and widely used.

2.2.1 Method 2: minimizing shovel idle time

The Method 2 algorithm for the problem of the dynamic dispatch of trucks in open-pit mining is a heuristic algorithm based on minimizing shovel idle time. The idea of the algorithm is presented by White and Olson (1986). Although the main idea follows the m-trucks-to-1-shovel strategy, we apply it based on the 1-truck-to-n-shovels strategy. In this method, the path that requires a truck soonest will be the best path (shovel) for assigning the truck. Interested readers are encouraged to consult the primary reference for more details. It is generally formulated by the following:

$$X_{[\mathrm{d,\ s}]} = L_{[\mathrm{d,\ s}]} + \frac{H_{[\mathrm{d,\ s}]}}{P_{[\mathrm{d,\ s}]}} - T_{[\mathrm{d,\ s}]} \tag{9}$$

where$X_{[\mathrm{d,\ s}]}$ The expected shovel idle time for the path from the current destination d to shovel s.$L_{[\mathrm{d,\ s}]}$ The time the last truck was allocated to the path from the current destination d to shovel s.$H_{[\mathrm{d,\ s}]}$ The haulage allocation at time $L_{[\mathrm{d,\ s}]}$ $P_{[\mathrm{d,\ s}]}$ The path flow rate from the current destination d to shovel s.$T_{[\mathrm{d,\ s}]}$ The travel time from the current destination d to shovel s.The neediest path is defined as:

$$TargetPath = Min(X_{[\mathrm{d,\ s}]}) \tag{10}$$

2.2.2 Method 3: minimizing ratio variance

This method follows a dynamic dispatch heuristic approach addressed by Xi and Yegulalp (1993) to find the best allocation schedule for a dispatch requisition. In this approach, when a truck requests a new assignment, the algorithm allocates the truck to a shovel based on minimizing the variance of the deviation from optimal path assignments determined by the upper stage in a multi-stage approach. For more detailed studies on the algorithm, the readers can refer to Xi and Yegulalp (1993). The expected path's deviation from the current destination d to shovel s is formulated as follows:

$$Dev_{[\mathrm{d,\,s}]} = f_{[\mathrm{d,\,s}]} - F_{[\mathrm{d,\,s}]} \times \overline{F} \tag{11}$$

$$\bar{F} = \frac{\sum_{d}\sum_{s} f_{[\mathrm{d,s}]} / F_{[\mathrm{d,s}]}}{N_F} \tag{12}$$

where $f_{[\mathrm{d,\,s}]}$ The cumulative path flow rate from the current destination d to shovel s. $F_{[\mathrm{d,\,s}]}$ The optimal path flow rate from the current destination d to shovel s obtained by the fleet allocation planning. $\overline{F}$ The mean value of ratios $f_{[\mathrm{d,s}]}/F_{[\mathrm{d,s}]}$ N_F The number of $F_{[\mathrm{d,\,s}]}$The neediest path is obtained as follows:

$$TargetPath = Min(Dev_{[\mathrm{d,\,s}]}) \tag{13}$$

2.2.3 Method 4: minimizing the deviation from the fleet allocation plan

In this method, the dispatching policy is made considering the deviation of the ore and waste volume ratio from their optimal production defined by the fleet allocation plan. The objective of the algorithm is to assign the trucks to shovels in a manner that the quality and quantity goals are met in the mining operation. In other words, the grade control of ore material and the desired stripping ratio are included in the real-time assignments. The needy shovels are sorted based on a measure indicating by how much they are behind schedule on their production. Then, the truck dispatches to the path with the minimum deviation. The approach is formulated by Eqs. (14)–(18).

$$Dev_{[\mathrm{t,\,s}]} = EVR_{[t,\,\mathrm{s}]} - TRVR_{[s]} \tag{14}$$

$$EVR_{[t,\,\mathrm{s}]} = \begin{cases} \frac{Vol_{[s]}+C_{[t]}}{VolT_{ore[t]}}, & s \in s_o \\ \frac{Vol_{[s]}+C_{[t]}}{VolT_{waste[t]}}, & s \in s_w \end{cases} \tag{15}$$

$$TRVR_{[s]} = \begin{cases} \frac{RV_{[s_o]}}{\sum_{s_o} RV_{[s_o]}}, & s \in s_o \\ \frac{RV_{[s_w]}}{\sum_{s_w} RV_{[s_w]}}, & s \in s_w \end{cases} \tag{16}$$

$$VolT_{ore[t]} = \sum_{s_o} Vol_{[s_o]} + C_{[t]} \tag{17}$$

$$VolT_{waste[t]} = \sum_{s_w} Vol_{[s_w]} + C_{[t]} \tag{18}$$

where$Dev_{[\mathrm{t,\,s}]}$ The expected deviation for shovel $s.EVR_{[t,\,\mathrm{s}]}$ The estimated volume ratio of shovel s if the truck t is sent to it.$TRVR_{[s]}$ The total required volume ratio of shovel s.$Vol_{[s]}$ The current volume of material available at shovel $s.C_{[t]}$ The capacity of truck t.$VolT_{ore[t]}$ The total volume of ore material hauled in the mine if truck t sent a dispatch request.$VolT_{waste[t]}$ The total volume of waste material hauled in the mine if truck t sent a dispatch request.$RV_{[s_o]}$ The required volume of ore material available at ore shovel s_o as specified by the fleet allocation schedule.$RV_{[s_w]}$ The required volume of waste material available at ore shovel s_w as specified by the fleet allocation schedule.$Vol_{[s_o]}$ The current volume of ore material available at ore shovel $s_o Vol_{[s_w]}$ The current volume of waste material available at ore shovel s_wThe neediest path is determined as follows:

$$TargetShovel = Min\left(Dev_{[\mathrm{t,\,s}]}\right) \tag{20}$$

3 Integrated simulation–optimization framework

To evaluate the proposed mathematical model in the real operation system, it is required to integrate simulator and optimizer software tools. Simulation refers to understanding the actual behavior of the transportation process in the truck-shovel system over time, while the optimization model seeks to find the best solution from a given solution space with regard to defined criteria. A variety of dedicated and highly specialized optimization and simulation software tools have appeared on the market over the past two decades. Among these tools, GAMS software is one of the most widely used optimization software packages. Meanwhile, Arena® is a general-purpose software tool for discrete event simulation (DES) and automation. The integrated framework consists of the optimization model developed in the GAMS software, the simulation model developed in the Arena software, and the Excel input file containing all required information to implement the proposed methodology in the mining operation. The integration and combined use of these commercial software tools are not officially supported. As shown in Fig. 4, Microsoft Visual Basic for Application (VBA) is a powerful development in technology that combines three software applications, including GAMS®, Arena®, and Excel®. By starting the simulation, Arena reads the historical data from the Excel file. The optimization model is called through VBA every time a truck requests an allocation. The optimal solutions obtained from the model are written in the same Excel file and read by the Arena® to provide real-time truck dispatching decisions. Therefore, by using Arena VBA, a customized integrated simulation–optimization method can be conducted that is both dynamic and flexible. Therefore, when a truck requests a new assignment, this leads to a timely solution.

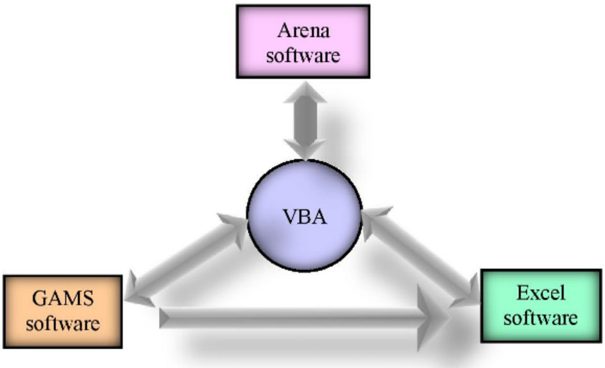

Fig. 4 Overall structure of the simulation–optimization technique developed in this study

4 Case study at a copper mine

The present case study refers to a real open-pit mine, Sungun copper mine, located in the northwest part of Iran country. To evaluate the developed framework in the case study, two types of data are required, the short-term production plan and the operational data. In this study, the short-term production schedule information used in the mine is considered. The hauling process of materials is being handled by a truck and shovel material handling system. Three companies, Mobin, Ahan-Ajin, and Noavarn, are separately in charge of the material transportation process. In this research, the Ahan-Ajin company's operational data were collected for analyzing the proposed methodologies. The mining operation comprises 4 loading points, corresponding to 4 loaders; 2 discharge points, defined according to the type of material as ore and waste; and a fleet of 19 haul trucks. The average capacity of the crusher is limited to 9000 tons per shift. There are two types within the set of 19 haul trucks; 15 correspond to type A with a nominal capacity of 80 tons and 4 to type B with a nominal capacity of 24 tons. All the hauling fleets in the mine are compatible to be loaded by any loading equipment. Table 1 summarizes the operating fleet data in the system for the target shift. The average grade of copper ore at loading points 3 and 4 are 0.75

Table 1 The operating fleet data summary in the given shift

Closed system	Loading point	Destination	Distance (m)	Available material	Loader type	Truck type	Number of trucks
1	1	Waste dump	2500	97,000	KOM-PC2000 (L_1)	KOM-HD758	5
2	2	Waste dump	3650	80,000	KOM-PC1250 (L_2)	KOM-HD758	7
3	3	Crusher	2100	15,000	KOM-PC1250 (L_3)	KOM-HD758	3
4	4	Crusher	3150	3000	KOM-PC1250 (L_4)	KOM-HD325	4

Table 2 Production rates of the loaders (t/shift)

Loader	Minimum production rate	Maximum production rate
L_1	5000	10,000
L_2	3800	7700
L_3	2800	7700
L_4	1350	7700

and 0.42%, respectively. According to the short-term production plan of the mine, the required stripping ratio to handle the extraction is in the range of 2 to 3. The minimum and maximum production rates of the loaders are also reported in Table 2 in the target shift. All trucks have the same speed due to the truck-bunching problem. In this mine, overtaking is banned. All trucks start the shift empty at a mining face. The mine operates for three 19.5-h shifts a day. In the case study, the dispatching method currently used by the mine is the fixed allocation and called truck locks, which means each truck is dedicated to a special shovel. The actual strategy not only does not try to satisfy the production requirements but also does not minimize the fleet waiting time. It is therefore important to provide better decision rules for the real-time truck dispatching in the mine.

The time study was carried out for an operational shift. Given the data set, there are eight sequential truck transportation activities, such as traveling, waiting, spotting, loading, hauling, queuing, maneuvering, and dumping. The collected truck cycle time data were analyzed with the Input Analyzer of Arena® software to generate the appropriate expressions for the sample values. The best-fitted distributions for the cyclic activities of the trucks were fitted on the historical data using operational data of the mine. The goodness-of-fit tests were estimated using the Kolmogorov–Smirnov and the Chi-Square tests with a 5% significance level. Table 3 presents the best-fit distribution models for the cyclic activities of the trucks.

5 Results and discussions

To evaluate the performance of the multi-objective truck dispatching model and the presented heuristic algorithms, the simulation experiments use the Arena® simulation software package in three steps are conducted. The first step is a validated simulation model of the case study to represent the operating states of the loading and hauling equipment system. In the second step, the mathematical dispatch model is applied using an integrated simulation–optimization technique. The next step is related to the implementation of the presented heuristic algorithms for real-time decisions. The most important key performance indicators (KPIs), based on which the performance of the approaches is evaluated, include delivered material tonnage to different destinations, truck cycle time, the total waiting time of equipment, queue lengths, utilization of equipment, and system utilization. Each simulation model was set up with 100 replications based on the required half-width for a shift of 6.5 h.

Table 3 The best-fit distribution models for the cyclic activities of the trucks

Data set	Destination	Loader	Hauler	Best fitted distribution
Spotting time	–	KOM-PC2000 KOM-PC1250	KOM-HD 758	UNIF (0.15, 0.22)
			KOM- HD 325	TRIA (0.11, 0.154, 0.18)
Loading time	–	KOM-PC2000	KOM-HD 758	TRIA (2.78, 3.37, 3.47)
		KOM-PC1250	KOM- HD 325	TRIA (1.26, 1.49, 1.56)
		KOM-PC2000	KOM- HD 325	TRIA (0.91, 1.23, 1.33)
		KOM-PC1250	KOM-HD 758	4.4 + 0.59 * BETA (3.46, 1.93)
Hauling time	Waste dump	KOM-PC2000 (S_1)	KOM-HD 758, KOM-HD 325	NORM (13.1, 0.579)
	Waste dump	KOM-PC1250 (S_2)		TRIA (14.7, 15.9, 16)
	Crusher	KOM-PC1250 (S_3)		7 + 3 * BETA (0.761, 0.645)
	Crusher	KOM-PC1250 (S_4)		14.3 + 0.701 * BETA (1.15, 0.496)
Maneuvering time	Waste dump Crusher	–	KOM-HD 758	TRIA (0.16, 0.235, 0.34)
			KOM- HD 325	NORM (0.191, 0.028)
Dumping time	Waste dump Crusher	–	KOM-HD 758	0.52 + LOGN (0.145, 0.0735)
			KOM- HD 325	TRIA (0.28, 0.344, 0.47)
Traveling time	Waste dump	KOM-PC2000 (S_1)	KOM-HD 758, KOM-HD 325	4.41 + GAMM (0.102, 5.38)
	Waste dump	KOM-PC1250 (S_2)		7.29 + 0.711 * BETA (1.03, 0.854)
	Crusher	KOM-PC2000 (S_1)		11.2 + 0.851 * BETA (1.21, 0.869)

Table 3 (continued)

Data set	Destination	Loader	Hauler	Best fitted distribution
	Crusher	KOM-PC1250 (S_2)		TRIA (15.2, 15.9, 16)
	Waste dump	KOM-PC1250 (S_3)		11.4 + ERLA (0.152, 4)
	Waste dump	KOM-PC1250 (S_4)		8 + 1 * BETA (1.21, 0.601)
	Crusher	KOM-PC1250 (S_3)		3.36 + 0.641 * BETA (1.32, 0.492)
	Crusher	KOM-PC1250 (S_4)		6.6 + ERLA (0.118, 4)
waiting time	–	KOM-PC1250 (S_2)	KOM-HD 758	3.67 + 0.91 * BETA (1.94, 3.04)

5.1 Model validation

The actual mine model was built based on the statistical expressions of the various cycle time elements defined in Sect. 4 to represent the actual characteristics of the loading and hauling operation. To validate the model and ensure that the model truly indicates the real system, truck cycle time as an important KPI was considered first. The results in Table 4 demonstrate that the simulation model is valid to a good extent and can reflect the accuracy of the fitted probability distributions on the cycle time parameters. Moreover, in the real operational system, the average waiting time of the trucks lining up in front of the mining face 2 is about 4 min, with a median of 2 trucks. Figure 5 confirms this result during the shift using the simulation model in the considered open-pit operation. Considering the other KPIs, the tonnage of material hauled from each mining face to destinations, and the total production of the mine for both simulated and real data are indicated in Figs. 6 and 7, respectively. All of these results state that the simulation model is in good agreement with the actual values in

Table 4 Comparison of truck cycle times recorded versus simulated results in the target shift (min)

Closed system	Rec			Sim			
	Mean	Min.	Max.	Mean	Half-width	Min.	Max.
1	22.4	20.3	23.7	22.3	0.02	20	24.3
2	32.5	29.9	33.7	32.9	0.01	28.6	34.4
3	18.3	16.5	19.9	18.8	0.03	16.6	22.3
4	24	23.4	24.6	24	0.01	23.2	25.3

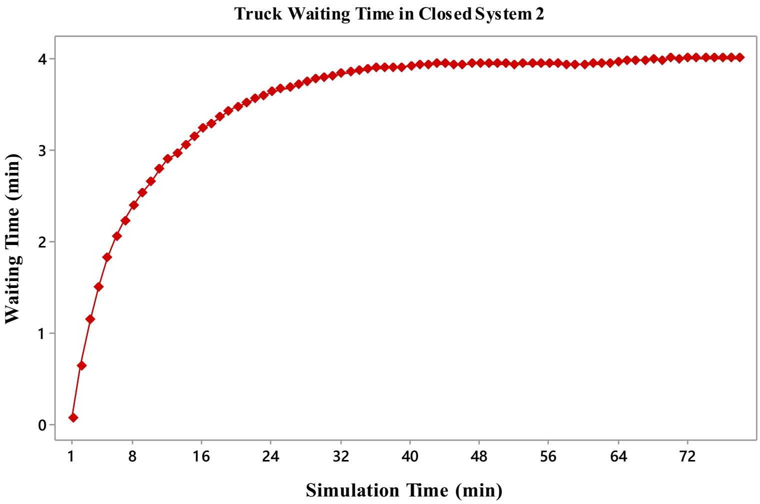

Fig. 5 Truck waiting time in the mining face 2 over time during the shift

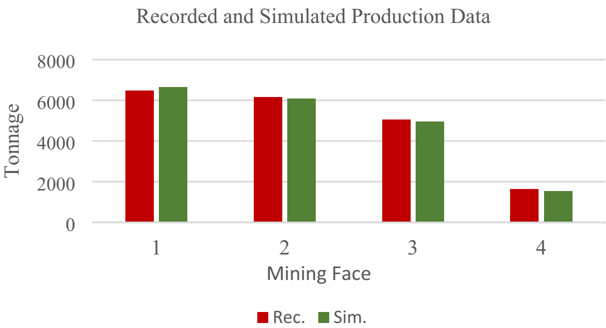

Fig. 6 Comparison of actual production data for each mining face with simulation results over the shift

Fig. 7 Comparison of the total mine production with the simulation results over the shift

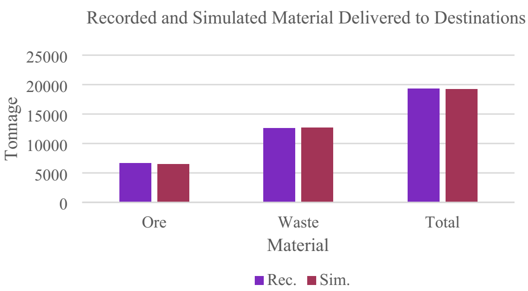

Table 5 Average utilization of the fleet and corresponding match factor in the case study

Closed system	Loader Util. (%)	Truck Util. (%)	Match factor
1	71	100	0.7
2	100	98	1.01
3	77	100	0.8
4	24	100	0.2

the real system, and this subsequently results in the model's excellent performance to mimic the real behavior of the mining operating system. Therefore, this model can be used as a basis for the implementation of the other proposed methods. In order to evaluate the performance of the real system, the average utilization of the fleet is examined. Moreover, the match factor is used to determine the efficiency of the trucks and shovels. The match factor is the ratio of the loader productivity to the truck productivity, which is formulated by Morgan and Peterson as:

$$MatchFactor = \frac{(\text{number of trucks}) \times (\text{loading time})}{(\text{number of loaders}) \times (\text{cycle time})} \tag{21}$$

The efficiency of the fleets is calculated using Eq. (21) in each closed system. If this factor is equal to one, the idle time of trucks and loaders is zero. If it is bigger than one, there are truck waiting times. If the match factor is smaller than one, there are shovel idle times in the operation. Table 5 presents the average fleet utilization and the related match factor in each closed system of the mining operation. As it can be seen from the table, the match factor in all closed systems except closed system 2 is less than one displaying an under-trucked operation with loader idle times. These results imply that the insufficient size of trucks was operated without considering the performance of the mining operation. Therefore, it is required to improve the efficiencies of the fleet in the mine to achieve the high performance of the system.

Thus, the first stage of the study consists of experiments to define the optimal number of trucks in the mining operation. To do so, different scenarios are generated using Arena Process Analyzer to determine the optimal number of trucks in the mine. The process starts to build scenarios by increasing the number of trucks from the current number in the consecutive scenarios. The scenario, in which the production requirements with the highest fleet utilization are met is considered the optimal scenario. Table 6 indicates the optimum number of the trucks using the scenario-based method. The results show that the operation requires 18 Komatsu HD785 trucks and 6 Komatsu HD325 trucks to meet the production requirements and increase the average fleet utilization, in which 7 Komatsu HD785 trucks are allocated to closed system 1, 7 Komatsu HD785 trucks to closed system 2, 4 Komatsu HD785 trucks to closed system 3, and 6 Komatsu HD325 trucks to closed system 4. The number of the trucks calculated in closed system 2 remains the same due to having a perfect match factor. There is no denying the fact that increasing the number of trucks in an under-trucked system leads to increasing fleet utilization and subsequently improving the match factor. As

Table 6 Simulation results of the scenario-based method in the given shift

Closed system	KOM- HD 758	KOM- HD 325	A match factor ratio	Truck util	Loader util
1	7	–	0.99	99	95
2	7	–	1.01	98	99
3	4	–	1.03	99	98
4	–	6	0.4	100	35

presented in Table 6, what is surprising is that the average loader utilization in closed system 4 is significantly low. This result is due to the lack of the available material at mining face 4, leading to loader idle times.

To deal with the upper stage of the proposed framework, the MILGP model was programmed in the GAMS software environment version 25.1.3 and solved with the CPLEX solver in a laptop computer with Win 7 operating system and 2.7 GHz Intel 7 Core CPU and 12 GB RAM. The computational time required to find the optimal solutions were obtained in less than a minute with a zero percent gap from the optimum solution. The results of the model, including the shovel allocations, the empty and loaded truck trips, and the paths' production rates, are provided in Table 7. The paths' production rates are applied as guidelines in the proposed dispatching methods to assign the trucks to the shovels.

In the next step, the presented dispatch algorithms were implemented considering the results of the first and second stages in the mining operation for the target shift. The following results and discussions focus on the analysis and comparison of the algorithms by the defined KPIs.

The delivered material tonnage to the proper destinations is one of the key KPIs that shows the performance of the presented approaches. Figure 8 demonstrates the impact of different methods on the system performance, by showing the total material (ore and waste) produced per shift. As shown in this figure, changing the dispatch strategy has a substantial impact on the results of delivered material tonnage. According to the Sungun copper mine dispatch plan, a total of 19,312 tons of materials (12,640

Table 7 Results of the truck and shovel allocation stage in the target shift

Mining face	1	2	3	4
Loader allocation	L_1	L_3	L_2	L_4
Optimal production (ton)	9000	7000	6640	1360
Loaded and empty truck trips	111 (KOM- HD 758) 5 (KOM- HD 325)	86 (KOM- HD 758) 5 (KOM- HD 325)	83 (KOM- HD 758)	17 (KOM- HD 758)

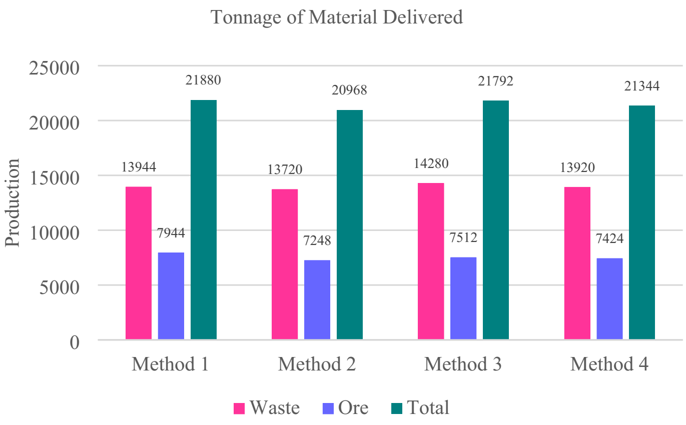

Fig. 8 The total materials produced using the presented methods in the target shift

tons of ore materials and 6672 tons of waste rocks) was produced in this shift. The comparison of the results obtained from the methods indicates that the productivity of the system using Method 1, Method 2, Method 3, and Method 4 increases by 13.3%, 8.5%, 12.8%, and 10.5%, respectively, compared to the current mine dispatch plan strategy. Therefore, the performance of the developed methods is optimal against the mine's strategy. In addition, the solutions yield good results in terms of the amount of the delivered materials, which are close to the optimal production of the truck-shovel allocation plan. Figure 8 shows that the total production of Method 1 and Method 3 are almost equivalent, while the amount of the ore and waste material produced in the mining operation from the two methods is different. By applying Method 1, the dispatching decisions are made so that the ore loaders are receiving the required trucks to feed the crusher and the waste loaders are receiving enough trucks to meet the stripping ratio requirement. However, in Method 3, the dispatch decision-making is based on sending more trucks to the waste loaders than ore loaders. The amount of the transported materials in Method 2 and Method 4 is about 4.3% and 2.5% lower than in Method 1. The waste material produced using Method 1, Method 2, and Method 4 is nearly the same. However, the ore tonnage produced in Method 2 is 8.7% lower than in Method 1. Moreover, when Method 4 is considered, the delivered ore tonnage is reduced by 6.5% compared to Method 1. Method 1 has the least deviation (0.7%) from the desired production of ore destination with the proposed model. By contrast, the highest deviations (9.4%) from the optimal ore production can be observed in Method 2. These results illustrate that Method 1 has the potential to be used for meeting the optimal production of the ore destination. The better results in the dispatch scheduling solution provided by Method 1 (Mathematical model) are mainly due to two main reasons: (1) the mathematical models find the best optimal solution in solving optimization problems, and (2) Multi-objective models a good tool to meet the production requirements in the mine operation level, which deal with multiple competing and conflicting objectives. Furthermore, among the heuristic algorithms, Method 3 is slightly outperformed compared to the other heuristics. This is likely owing to the strategy of the developed algorithm, which leads to good outcomes. Therefore, the findings indicate that replacing the mine dispatch plan with the multi-objective dispatching model will help to meet the production requirements in the operating shift.

Another aspect of the results of the methods to be discussed is the performance and efficiency of the equipment in the mining operation, as it has a direct impact on mine profitability and operational costs. Table 8 compares the results of the fleet utilization obtained from the developed dispatch strategies to display the effect of different strategies in the mining operation. The differences in the results are mainly due to the fact that each method has a specific dispatching strategy. These results confirm the effect of the increased loader loading time on the utilization of the loaders, especially for the loaders which have sent more trucks during the shift. In all methods, the trucks are utilized more than 95% of their available time because of an appropriate truck size in the mining operation. Moreover, as explained earlier, the poor performance of loader 4 stems from a lack of the available ore materials at the mining face 4. The low utilization suggests a substantial amount of idling of the loaders at the loading sites. According to this table, on average, the operation implementing Method 1 produces higher utilization of the loaders over the shift. It is obvious that the increase in the utilization of the fleet leads to increased production of the operation. That is why the amount of transported material in Method 1 is higher than in the other algorithms. Considering the methods' productivity, it is not unexpected that the lowest performance of the loaders is observed in Method 2. Thus, it can be concluded that the real-time dispatch decisions using Method 1 consider a more effective and efficient usage of the resources during the operation and can be helpful in the actual mining operation.

The number of trucks waiting in a queue at a loading point is another KPI that helps to evaluate the operational approaches in any mining system. Figure 9 represents the average queue lengths for the considered methods. As the amount of materials produced is largely dependent on the utilization of loaders, and subsequently truck queues at loaders, a high level of the truck queues using Method 2 can be inferred, as confirmed in Fig. 9. Considering the results of the methods, Methods 1, 3, and 4 have almost the same lineup in the queue of the loaders with a medium of 4 trucks in the considered shift of the mine operation. However, Method 1 helps the operation to remove more materials compared to the other algorithms during the shift.

Table 8 Comparison of the fleet utilization in the mining operation using presented methods for the shift

Face	Loader Util. (%)				Truck Util. (%)			
	Method 1	Method 2	Method 3	Method 4	Method 1	Method 2	Method 3	Method 4
1	96	50	99	70	96	100	99	100
2	79	99	78	99	98	94	98	96
3	95	86	89	87	98	95	97	95
4	21	21	21	21	100	100	100	100

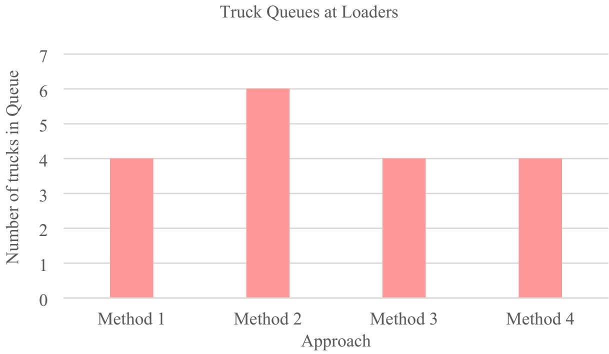

Fig. 9 Average truck queues at loaders using the presented methods in the target shift

Generally, the proposed optimization model provides several advantages compared to the other heuristic algorithms as follows:

- Improving the performance of the fleets.
- Increasing the productivity of the system.
- Meeting the optimal production requirements in each path.
- Reducing the loader idle times.
- Receiving required trucks by the ore loaders to feed the crusher and receiving enough trucks by the waste loaders to meet the stripping ratio requirements.

6 Conclusions

One significant problem in the profitability of open-pit mines is an efficient truck dispatching system, as it can result in substantial cost reductions. This work presents a multi-stage dispatching approach based on the 1-truck-to-n-shovels strategy for open-pit mining operations to increase the utilization of the mining equipment. In the first stage, a scenario-based method is presented to determine the optimum haulage fleet size in the mining operation. In the next step, a multi-objective allocation model is chosen for the upper stage in the presented multi-stage decision-making for the real-time truck dispatching problem. The model is formulated based upon a MILGP model to optimize the operations considering four desired goals: (1) maximize production, (2) minimize deviations in head grade, (3) minimize deviations in tonnage to the ore destinations, and (4) minimize fuel consumption of mining trucks. Then, a novel multi-objective model is developed for real-time truck dispatching decisions. The model considers as objectives the minimization of fleet waiting times and deviations from the path production requirements defined by the allocation planning. The aim is to find an allocation for the haul trucks that best meet the allocation schedule. To evaluate the performance of the proposed model, three different heuristic methods, including criteria of minimizing shovel idle time, minimizing ratio variance, and minimizing the deviations from the fleet allocation plan, are considered. The effectiveness of each method is demonstrated by embedding them in a discrete-event simulation of a copper ore mine case study. By changing the dispatch policy of the mine, the results suggest that the proposed model increases equipment utilization and the productivity of the mine compared to the heuristic algorithms. As future work, we intend to develop the proposed model based on the m-trucks-to-n-shovels strategy to compare the performance of the different strategies.

Declarations

Conflict of interest The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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